

# Statistical Tools and Measurements for PHY 105M Lab for Engineering Physics I

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## Comparing Measurements: T-Score

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\delta \bar{x}_1^2 + \delta \bar{x}_2^2}} \quad (1)$$

Intuitively this measures the difference in the mean in units of the uncertainty. Larger/smaller values of  $t$  mean a larger/smaller discrepancy between the values.

We will use a convention where:

$t < 1$  is considered *indistinguishable*

$1 < t < 3$  is considered *inconclusive*

$3 < t$  is considered *distinguishable*

**NOTE:** the language is "indistinguishable" not "the same". In science we falsify or fail to falsify models we don't "confirm" them.

## Propagation of Uncertainty

- Each uncertainty contributes to the total uncertainty proportional to the partial derivative (slope of  $f$  with respect to that variable).
- Multiple sources don't simply add but add "in quadrature", since they are as likely to cancel as add.
- We will always give you the needed derivatives.

$$\delta f = \sqrt{\sum_i \left( \frac{\delta f}{\delta x_i} \delta x_i \right)^2} \quad (2)$$

$$\delta f(x, y) = \sqrt{\left( \frac{\delta f}{\delta x} \delta x \right)^2 + \left( \frac{\delta f}{\delta y} \delta y \right)^2} \quad (3)$$

There are three steps to getting the propagation of uncertainty formula: **Terms, Variables, & Derivatives**.

$$\delta f(z_1, z_2, \dots, z_N) = \sqrt{\sum_{i=1}^N \left(\frac{\delta f}{\delta z_i} \delta z_i\right)^2} \quad (4)$$

## Terms

Start with the general propagation of uncertainty formula and *write out the same number of terms as you have uncertainties* (three in this example:  $\delta m, \delta v, \delta r$ )

$$\delta f(z_1, z_2, z_3) = \sqrt{\sum_{i=1}^3 \left(\frac{\delta f}{\delta z_i} \delta z_i\right)^2} \equiv \sqrt{\left(\frac{\delta f}{\delta z_1} \delta z_1\right)^2 + \left(\frac{\delta f}{\delta z_2} \delta z_2\right)^2 + \left(\frac{\delta f}{\delta z_3} \delta z_3\right)^2} \quad (5)$$

## Variables

Associate each variable with a  $z_i$  (order doesn't matter) and associate the function with  $f$

$$\begin{aligned} z_1 &\mapsto m \\ z_2 &\mapsto v \\ z_3 &\mapsto r \\ f &\mapsto F_c \end{aligned}$$

Then directly substitute into the formula from the *Terms* step.

$$\delta F_c(m, v, r) = \sqrt{\left(\frac{\delta F_c}{\delta m} \delta m\right)^2 + \left(\frac{\delta F_c}{\delta v} \delta v\right)^2 + \left(\frac{\delta F_c}{\delta r} \delta r\right)^2} \quad (6)$$

## Derivatives

Substitute in the derivatives from equation (1) to get the **correct formula**

$$\sqrt{\left(\frac{v^2}{r} \delta m\right)^2 + \left(\frac{2mv}{r} \delta v\right)^2 + \left(-\frac{mv^2}{r^2} \delta r\right)^2}$$